

From topology to combinatorics via computability theory

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Recap

Theorem. If $P_0, \dots, P_{k-1}$ admit cone avoidance then there is an ω -model of $\text{RCA}_0 + P_0, \dots, O_{k-1} + \neg \text{ACA}_0$.

COH admits cone avoidance.

- ▶ We used Mathias forcing with computable reservoirs.
 - ▶ Looked for e-splits to force a disagreement with a given $C \not\leq_T \emptyset$.
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RT_2^1 admits strong cone avoidance.

Lecture 4

Strong cone avoidance of RT_2^1

Theorem (Dzhafarov & Jockusch). RT_2^1 admits strong cone avoidance.

We again fix $C \not\leq_T \emptyset$ and $B \not\leq_T C$. For ease of notation, take $B = \emptyset$.
Let $d : \omega \rightarrow 2$ be arbitrary and write $A_i = \{x : d(x) = i\}$.

Seeking a contradiction, suppose neither A_i has an infinite subset $G \not\leq_T C$.

We build **two sets** G_0 and G_1 that meet the following for all n, e_0, e_1 :

- ▶ $\mathcal{Q}_n : |G_0| \geq n \wedge |G_1| \geq n$;
- ▶ $\mathcal{R} : G_0 \subseteq A_0 \wedge G_1 \subseteq A_1$;
- ▶ $\mathcal{S}_{e_0, e_1} : \Phi_{e_0}^{G_0} \neq C \vee \Phi_{e_1}^{G_1} \neq C$.

(NB this is a disjunction!)

$\mathcal{S}_{e_0, e_1}$ implies $(\forall e)[\Phi_e^{G_0} \neq C]$ or $(\forall e)[\Phi_e^{G_1} \neq C]$. So one of G_0 or G_1 works.

(Why?)

Double Mathias conditions

We force with conditions that are triples (D_0, D_1, I) such that:

- ▶ (D_0, I) and (D_1, I) are Mathias conditions;
- ▶ $I \not\leq_T C_i$;
- ▶ $D_0 \subseteq A_0$ and $D_1 \subseteq A_1$.

A sufficiently generic is thus a pair $\langle G_0, G_1 \rangle$ such that $G_0 \subseteq A_0$ and $G_1 \subseteq A_1$, which is how we satisfy $\mathcal{R}$.

To satisfy $\mathcal{Q}_n$, let (D_0, D_1, I) be any condition.

- ▶ If $I \cap A_i$ is finite for either i then $I \subseteq^* A_{1-i}$, which cannot be since $I \not\leq_T C$.
- ▶ So let F_i consist of n elements of $I \cap A_i$ and pass to the extension $(D_0 \cup F_0, D_1 \cup F_1, I \setminus \max(F_0 \cup F_1))$ of (D_0, D_1, I) .

Cone avoiding basis theorem

Recall:

- ▶ $\mathcal{C} \subseteq 2^\omega$ is a Π_1^0 class if $\mathcal{C} = \{X \in 2^\omega : (\forall k) R(k, X \upharpoonright k)\}$ for some $R \leq_T \emptyset$;
- ▶ a **basis for Π_1^0 classes** is a collection $\mathcal{B}$ of sets such that if $\mathcal{C} \subseteq 2^\omega$ is any nonempty Π_1^0 class then $\mathcal{C} \cap \mathcal{B} \neq \emptyset$.

Theorem (Jockusch and Soare). The following are bases for Π_1^0 classes:

- ▶ the class of all low sets; (low basis theorem)
- ▶ the class of all sets of hyperimmune-free degree; (hyperimmune-free basis theorem)
- ▶ for every $C \not\leq_T \emptyset$, the class of all $X \not\leq_T C$. (cone avoidance basis theorem)

Our interest is in the cone avoidance basis theorem.

Satisfying strong cone avoidance

$$\mathcal{S}_{e_0, e_1} : \Phi_{e_0}^{G_0} \neq C \vee \Phi_{e_1}^{G_1} \neq C.$$

Let (D_0, D_1, I) be any condition.

We claim there is an extension (E_0, E_1, J) of (D_0, D_1, I) and an $i < 2$ such that if G_i is any set satisfying (E_i, J) then $\Phi_{e_i}^{G_i} \neq C$.

Let $\mathcal{C}$ be the class of all sets $X = \langle X_0, X_1 \rangle$ such that $X_0 \sqcup X_1 = I$ and

$$(\forall i < 2)(\forall F_0, F_1 \subseteq X_i \text{ finite})(\forall x)[\Phi_{e_i}^{D_i \cup F_0}(x) \simeq \Phi_{e_i}^{D_i \cup F_1}(x)].$$

(There are no e_i -splits over D_i in X_i .)

Note that $\mathcal{C}$ is a $\Pi_1^0(I)$ class.

Satisfying strong cone avoidance

$\mathcal{C}$ is the class of all sets $X = \langle X_0, X_1 \rangle$ such that $X_0 \sqcup X_1 = I$ and

$$(\forall i < 2)(\forall F_0, F_1 \subseteq X_i \text{ finite})(\forall x)[\Phi_{e_i}^{D_i \cup F_0}(x) \simeq \Phi_{e_i}^{D_i \cup F_1}(x)].$$

Case 1: $\mathcal{C} \neq \emptyset$.

By the cone avoidance basis theorem, [relativized to \$I\$](#) , there exists $X = \langle X_0, X_1 \rangle \in \mathcal{C}$ such that $I \oplus X \not\leq_T C$.

Since $X_0 \sqcup X_1 = I$ and I is infinite, X_i is infinite for some i . Pass to the extension $(E_0, E_1, J) = (D_0, D_1, X_i)$.

Now if G_i is any set that satisfies (D_i, X_i) then $\Phi_{e_i}^{G_i}$ cannot be total.

(Same argument as earlier.)

Satisfying strong cone avoidance

$\mathcal{C}$ is the class of all sets $X = \langle X_0, X_1 \rangle$ such that $X_0 \sqcup X_1 = I$ and

$$(\forall i < 2)(\forall F_0, F_1 \subseteq X_i \text{ finite})(\forall x)[\Phi_{e_i}^{D_i \cup F_0}(x) \simeq \Phi_{e_i}^{D_i \cup F_1}(x)].$$

Case 2: $\mathcal{C} = \emptyset$.

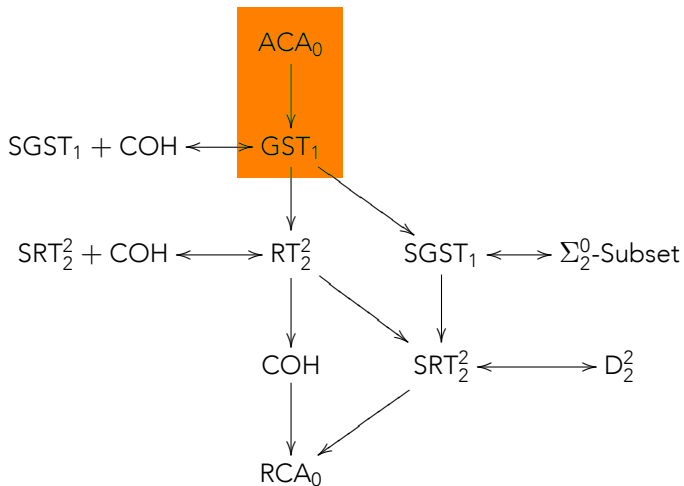
By compactness of 2^ω , we can fix k such that for any $X_0 \sqcup X_1 = I$ we have

$$(\exists i < 2)(\exists F_0, F_1 \subseteq X_i \upharpoonright k)(\exists x)[\Phi_{e_i}^{D_i \cup F_0}(x) \downarrow \neq \Phi_{e_i}^{D_i \cup F_1}(x) \downarrow].$$

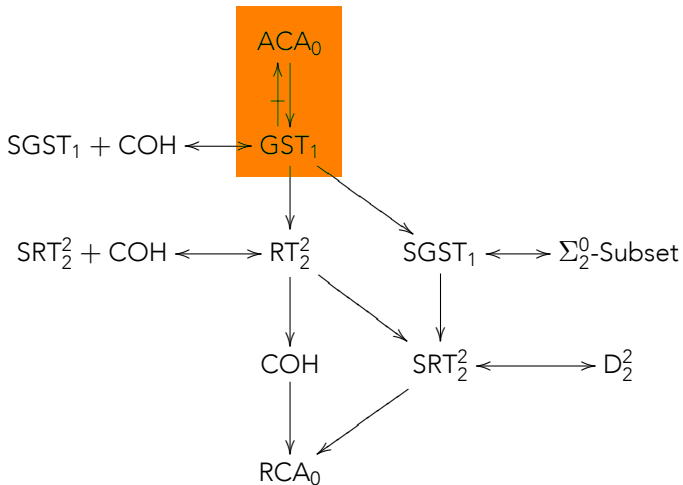
Apply this with $X_0 = I \cap A_0$ and $X_1 = I \cap A_1$ and get $i < 2$ and $F_0, F_1 \subseteq X_i \upharpoonright k$.

For some $j < 2$ then $\Phi_{e_i}^{D_i \cup F_j}(x) \neq C(x)$. Pass to the extension (E_0, E_1, J) where $\blacktriangleright E_i = D_i \cup F_j$ $\blacktriangleright E_{1-i} = D_{1-i}$ $\blacktriangleright$ and $J = I \setminus \varphi_{e_i}^{D_i \cup F_j}(x)$.

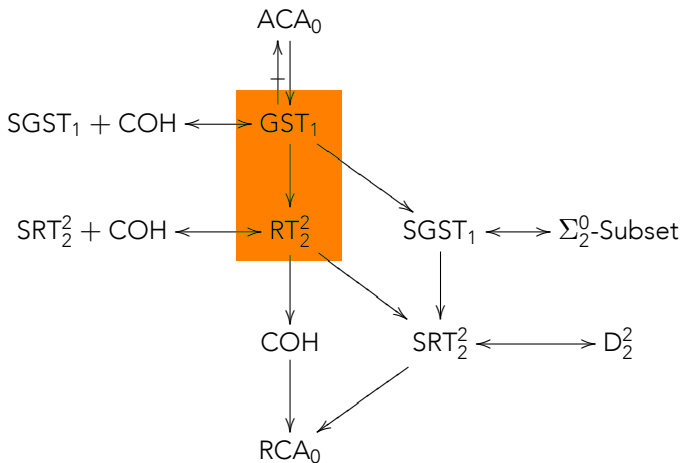
Where we are



Where we are



Where we are



Designing a separation property

Suppose P and Q are problems and we want an ω -model of $P + \neg Q$.

To do this, we can look for a **computability-theoretic property** $\mathcal{C}$ such that

- ▶ P **preserves** $\mathcal{C}$;

(Every instance $X \in \mathcal{C}$ has a solution Y s.t. $X \oplus Y \in \mathcal{C}$.)

- ▶ Q **does not preserve** $\mathcal{C}$.

(There is an instance $X \in \mathcal{C}$ s.t. $X \oplus Y \notin \mathcal{C}$ for all solutions Y .)

So long as $\mathcal{C}$ is downwards closed under $\leq_T$, we can **iterate and dovetail** to obtain the desired ω -model exactly like in the cone avoidance metatheorem.

- ▶ $\mathcal{C} = \{X : X \leq_T \emptyset\}$ is preserved only by computably true problems.
- ▶ $\mathcal{C} = \{X : X \not\leq_T \emptyset'\}$ is preserved by Σ_2^0 -Subset but not ACA_0 .

Designing a separation property

A property $\mathcal{C}$ preserved by Δ_2^0 -Subset but not Σ_2^0 -Subset.

We want:

- ▶ a Σ_2^0 set S such that no infinite subset of S or $\bar{S}$ is in $\mathcal{C}$; (a single S)
- ▶ if $X \in \mathcal{C}$ then for every $\Delta_2^0(X)$ set A there is an infinite subset G of A or $\bar{A}$ such that $X \oplus G \in \mathcal{C}$. (one G for every possible A)

So at minimum, $\mathcal{C}$ must imply not computing an infinite subset of S or $\bar{S}$.
(S and $\bar{S}$ must be X -immune for every $X \in \mathcal{C}$)

If $\mathcal{C}$ says more than this, then

- ▶ we have more to work with from a given $X \in \mathcal{C}$;
- ▶ but also more that we need to make G satisfy to guarantee $X \oplus G \in \mathcal{C}$.

So designing $\mathcal{C}$ is a careful balancing act.

Hyperimmune sets and PA degrees

Fix sets X and Y . Recall that:

- ▶ X is **Y -hyperimmune** if there is no Y -computable sequence $x_0 < x_1 < \dots$ such that $[x_n, x_{n+1}) \cap X \neq \emptyset$ for all n . (No Y -computable strong array for X)
- ▶ X is **Y -hyperimmune-free** if it computes no Y -hyperimmune set.
- ▶ X is **PA relative to Y** (written $X \gg Y$) if every nonempty $\Pi_1^0(Y)$ class has an X -computable element.

Note: If X is Y -hyperimmune it is also Y -immune.

There is a Π_1^0 class $\mathcal{PA}$ such that for
 $\{\deg_T(X) : X \gg \emptyset\} = \{\deg_T(X) : X \in \mathcal{PA}\}.$

By the hyperimmune-free basis theorem, every nonempty Π_1^0 class has a hyperimmune-free member. So there exists a hyperimmune-free $X \gg \emptyset$.

The property we want

Theorem. There exists a Σ_2^0 set S for which there exist a low set $L_0 \gg \emptyset$ and a set $P_0 \gg \emptyset'$ such that S is L_0 -hyperimmune and $\bar{S}$ is P_0 -hyperimmune.

With the above S fixed, we let $\mathcal{C}$ consist of all X for which there exist

- ▶ a set $L \gg X$,
- ▶ a set $P \gg L'$

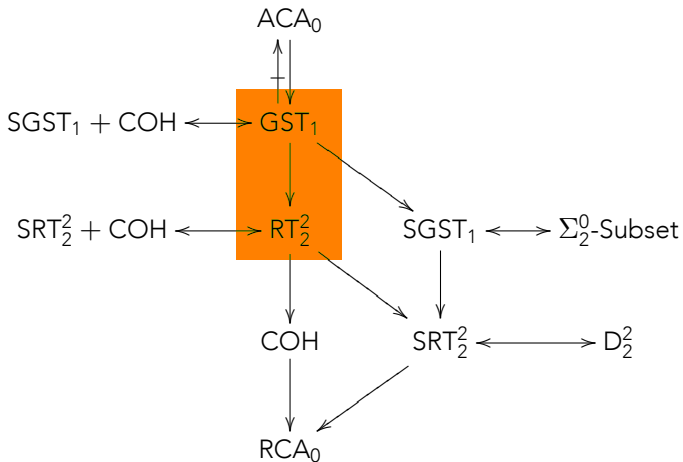
such that S is L -hyperimmune and $\bar{S}$ is P -hyperimmune.

Crucially: if $X \in \mathcal{C}$ then S and $\bar{S}$ are X -immune.

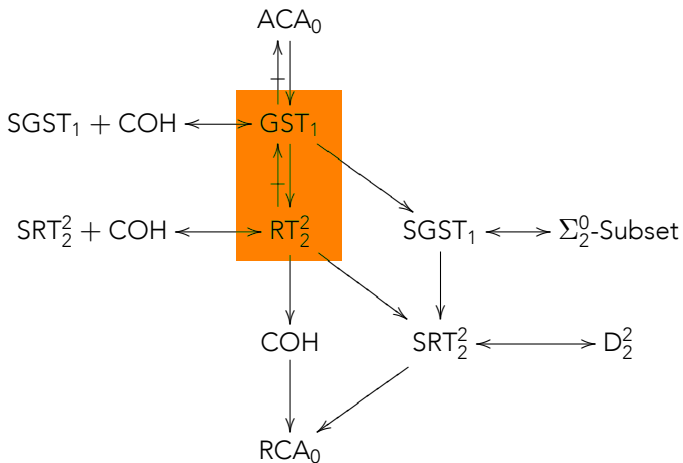
Thus S witnesses that Σ_2^0 -Subset does **not** preserve $\mathcal{C}$ because:

- ▶ $\emptyset \in \mathcal{C}$, as witnessed by $L = L_0$ and $P = P_0$;
- ▶ S is a computable instance of Σ_2^0 -Subset but no solution to S can be in $\mathcal{C}$.

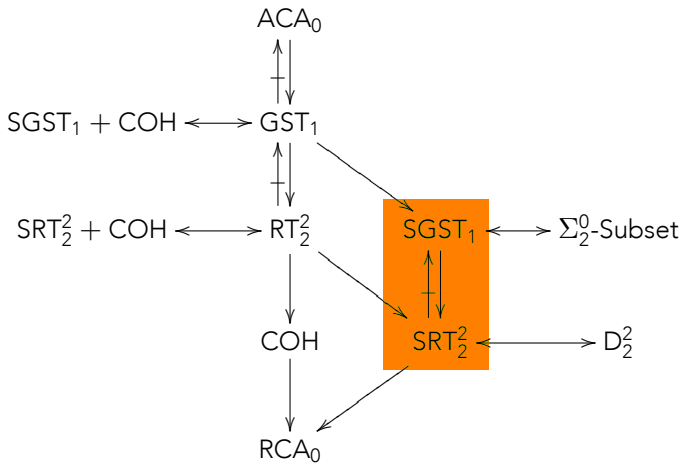
Where we are



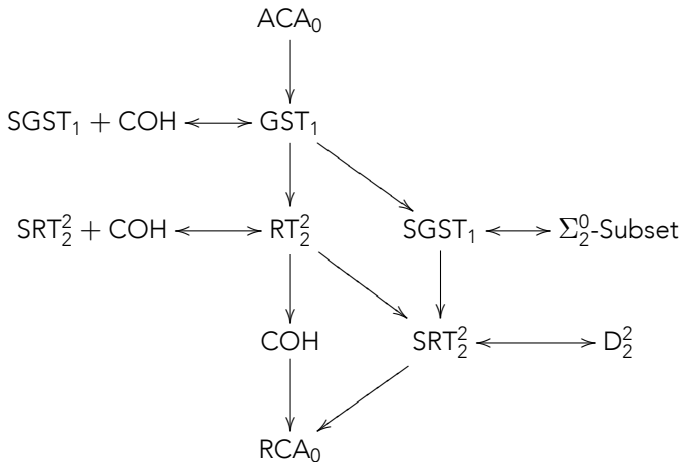
Where we are



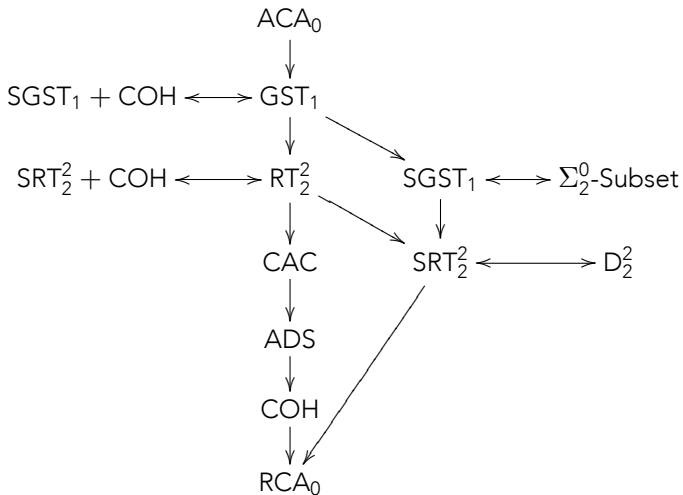
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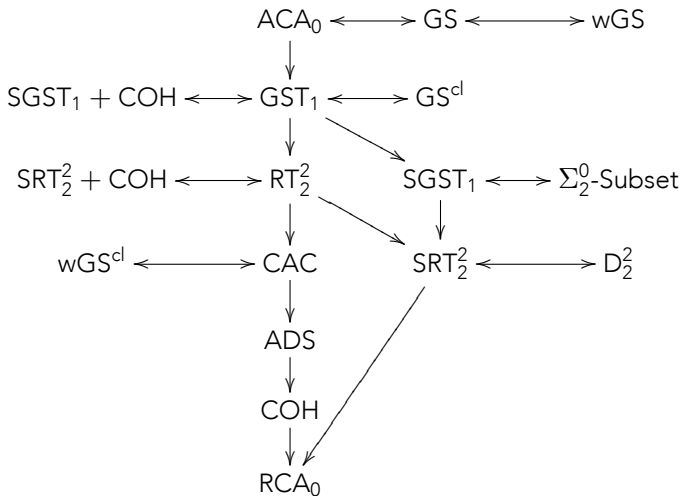
Summary



Summary



Summary



wGS^{cl}, revisited

Consider a CSC space X with its closure operator cl , given.

Using cl , we can form the relation $\trianglelefteq$ defined by $x \trianglelefteq y \leftrightarrow x \in \text{cl}(y)$.

If X is T_0 then $\trianglelefteq$ is a partial order, but in general it is only a **quasiorder**.

(We may have $x \trianglelefteq y$ and $y \trianglelefteq x$ for some $x \neq y$.)

By wGS^{cl}, X has an infinite subspace Y that is either indiscrete, has the initial segment topology, has the final segment topology, or is T_1 .

- ▶ If Y is indiscrete then Y is a $\trianglelefteq$ -equivalence class.
- ▶ If Y has the initial or final segment topology then Y is a $\trianglelefteq$ -descending or $\trianglelefteq$ -ascending sequence, respectively.
- ▶ If Y is T_1 then Y is a $\trianglelefteq$ -antichain.

wGS^{cl}, revisited

Definition (Benham).

QADAC: Every quasiorder $\leq_Q$ on $\mathbb{N}$ has an infinite equivalence class, ascending sequence, descending sequence, or equivalence class.

ADAC: Every partial order $\leq_P$ on $\mathbb{N}$ has an infinite ascending sequence, descending sequence, or antichain.

Theorem (Benham). Over RCA_0 , $\text{wGS}^{\text{cl}} \leftrightarrow \text{QADAC} \leftrightarrow \text{ADAC}$.

That $\text{wGS}^{\text{cl}} \leftrightarrow \text{QADAC}$ is by the previous slide. Clearly, $\text{QADAC} \rightarrow \text{ADAC}$.

Now fix a quasiorder $\leq_Q$. Assume there is no infinite $\leq_Q$ -equivalence class.

Since $\text{ADAC} \rightarrow \text{ADS} \rightarrow \text{B}\Sigma_2^0$, we can use ADAC to find an infinite set Y of $\leq_Q$ -representatives.

$$(\forall x)(\exists y > x)(\forall z \leq x)[z \not\leq_Q y \vee y \not\leq_Q z]$$

Now $\leq_Q$ restricted to Y is a partial order; apply ADAC.

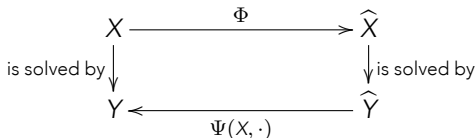
Weihrauch reducibility

Let P and Q be instance-solution problems.

P is **Weihrauch reducible** to Q , written $P \leq_w Q$, if there are Turing functionals Φ and Ψ such that

- ▶ if X is a P -instance then $\hat{X} = \Phi(X)$ is a Q -instance;
- ▶ if $\hat{Y}$ is a Q -solution to $\Phi(X)$ then $Y = \Psi(X, \hat{Y})$ is a P -solution to X .

So the following diagram “commutes”:



(Weihrauch 1992; Brattka; Gherardi and Marcone 2008; DDHMS 2016).

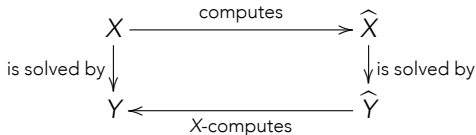
Computable reducibility

Let P and Q be instance-solution problems.

P is **computably reducible** to Q , written $P \leq_c Q$, if

- ▶ every P -instance X computes a Q -instance $\hat{X}$;
- ▶ if $\hat{Y}$ is a Q -solution to $\hat{X}$ then $X \oplus \hat{Y}$ computes a P -solution Y to X .

So the following diagram “commutes”:



(Dzhafarov 2015; Patey 2015/2016; Hirschfeldt and Jockusch 2016).

A closer look

Theorem (Benham). Over RCA_0 , $\text{wGS}^{\text{cl}} \leftrightarrow \text{QADAC} \leftrightarrow \text{ADAC}$.

Easy to check that $\text{wGS}^{\text{cl}} \equiv_W \text{QADAC}$.

Clearly, $\text{ADAC} \leq_W \text{QADAC}$.

The proof that $\text{ADAC} \rightarrow \text{QADAC}$ shows $\text{QADAC} \leq_c \text{ADAC}$, but it is **not** uniform: we asked whether or not there was an infinite equivalence class.

Theorem (Benham). $\text{QADAC} \not\leq_W \text{ADAC}$.

Earlier, we saw that $\text{wGS}^{\text{cl}} \leftrightarrow \text{CAC}$. Clearly, $\text{CAC} \leq_W \text{ADAC}$. But to get ADAC from CAC , we need **two applications** (first CAC , then ADS).

Theorem (Benham). $\text{ADAC} \not\leq_c \text{CAC}$.

Preservation of hyperimmunities

Definition (Patey). A problem P **preserves p among k hyperimmunities** if, given k many hyperimmune sets $X_0, \dots, X_{k-1}$, every computable P -instance has a solution Y such that at least p of the X_i are Y -hyperimmune.

Theorem (Benham). CAC preserves 2 among 3 hyperimmunities.

On the other hand, by a result of Patey there is a computable partial order $\leq_P$ such that $NS = \{x : x \text{ is not } \leq_P\text{-small}\}$, $NL = \{x : x \text{ is not } \leq_P\text{-large}\}$, and $NI = \{x : x \text{ is not } \leq_P\text{-isolated}\}$ are all hyperimmune.

Applying ADAC to $\leq_P$ we get either:

- ▶ an infinite ascending sequence;
- ▶ an infinite descending sequence;
- ▶ an infinite antichain;

$$\subseteq NL \cap NI$$

$$\subseteq NS \cap NI$$

$$\subseteq NL \cap NS$$

So ADAC does not preserve 2 among 3 hyperimmunities.